Venkata Lakshmi Parimala Pasupuleti

AI Engineer · Agentic AI · MCP Servers · LangGraph · LLM Evaluation Harnesses · Enterprise Automation · Python · Production AI

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Virginia, USA · Open to Remote

PROFESSIONAL SUMMARY

AI Engineer with 8+ years of production engineering and hands-on expertise in **agentic AI, LangGraph, MCP servers, LLM evaluation harnesses, and enterprise workflow automation** -- designing, deploying, and validating autonomous AI agents that perform complex multi-step reasoning reliably in production. Built **Leasel**: an 8-agent LangGraph enterprise automation pipeline (Triage → Retrieval → Configuration → Historical Incident → Structured Data → Reasoning → Validation → Human Escalation) with confidence-gated HITL and full production observability -- 93.6% precision across 1,240 enterprise documents. Built **AgentProbe**: a CI-integrated LLM evaluation harness with 5-metric LLM-as-judge scoring, scenario simulation, regression detection, and automated quality gates -- blocks failing agent runs before production deploy. Shipped Entrata's production AI pipeline across 1.6M financial records, eliminating 80% of manual operational workflows. M.S. Data Analytics Engineering, GMU (GPA 3.87, May 2026).

CORE COMPETENCIES

Agentic AI & Multi-Agent Systems | LangGraph & LangChain | MCP Servers & Custom Toolchains | LLM Evaluation Harnesses | CI/CD for LLM-Powered Systems | Enterprise Workflow Automation | RAG Pipeline Design | Production AI & Reliability | HITL & Agent Governance | Python & FastAPI | AWS / Azure / GCP | Prompt Engineering

EXPERIENCE

Blue Horse Digital

Sep 2026 – Present

AI/ML Engineer (Volunteer) · Python · ML Pipelines · Data Analysis

- Building ML models and agentic data pipelines for business intelligence; integrating AI components with back-end APIs and reporting layers.

Magizhs Tech Solution

May 2025 – Aug 2025

AI/ML Engineer Intern · LangGraph · Agentic Automation · RAG · Compliance Guardrails

- **Built a domain-specific agentic automation system** (LangGraph stateful graph: parse → plan → retrieve → validate → conditional routing) that translates natural-language requests into structured, validated outputs -- deployed with production-ready guardrails blocking non-compliant outputs before delivery.
- Implemented a **RAG compliance layer** over a regulatory corpus (FAA Part 107) with cited-source answers; engineered deterministic validation guardrails enforcing business rules at every stage -- same architectural pattern as enterprise sales agent automation.

George Mason University — Research Associate / Teaching Assistant

Aug 2024 – May 2026

AI/ML Research · Python · Statistical Modeling · Instruction

- Built predictive models (Python, R) achieving $R^2 = 0.977$ across 190+ countries; applied Isolation Forest anomaly detection for automated data integrity checks; designed analytics dashboards improving research comprehension 40% across two faculty cohorts.
- Mentored 600+ students across Python programming and database design (SQL, ER modeling, normalization).

Entrata (PropTech / Real Estate SaaS)

Sep 2018 – Aug 2024

Senior Software Engineer — Enterprise Automation · Production AI · Python · Agentic Workflows

- **Eliminated 80% of manual enterprise workflows** via Python/Bash automation pipelines -- automated annual income-limit imports (few thousand federal rows → 180K domain records), gross-rent bulk updates, and document-validation cycles across 100K+ housing units; zero compliance violations across 7 years of release cycles.
- **Shipped production AI pipeline across 1.6M financial records** (Python, PyTorch, OpenAI API) -- designed autonomous processing workflows with output validation gates, regression detection, and compliance enforcement; integrated AI outputs with enterprise APIs (REST, AWS S3) and downstream production systems at 99.5% uptime.
- **Built RabbitMQ async processing system** with idempotency guarantees for enterprise-scale workflow automation (thousands of tenant records); designed retry logic eliminating double-processing failures; owned the system through production incidents and annual regulatory release cycles.
- Optimized 200+ SQL queries (30% performance gain); led Docker containerization reducing microservices execution time 50%; conducted 300+ peer code reviews; mentored 5+ engineers on production standards and automated testing practices.

Entrata India (Db Xento Systems)

Jan 2017 – Sep 2018

Associate Software Engineer — Python · Backend · Data Pipelines · AWS

- Built compliance data pipelines (Python, PHP, PostgreSQL) processing 1.6M HUD/Tax Credit records; led AWS S3 cloud migration with 100% PHI compliance; contributed to API integrations and database architecture redesigns.

PORTFOLIO PROJECTS

AgentProbe — CI-Integrated LLM Evaluation Harness

Eval Harness · CI/CD for LLM · LangGraph

2025 · Python, FastAPI, LangGraph (3 agents), ChromaDB, OpenAI/Anthropic/Groq/Ollama · github.com/venkatalakshmiparimala/AgentProbe

- Built a **CI/CD evaluation harness for LLM-powered agent systems**: 15-endpoint FastAPI backend, **5-metric LLM-as-judge scoring** (factual accuracy, hallucination, retrieval, completeness, compliance), **scenario simulation** over golden datasets, **regression detection** gates that block production deploys when quality drops below threshold -- fully local, zero vendor dependency.
- Three LangGraph agents (Orchestrator, RAG Tracer, Auto Fix) handle multi-provider LLM routing (OpenAI, Anthropic, Groq, Ollama); React observability dashboard surfaces per-run regression drift; CI-integrated via GitHub Actions -- used to validate LeaseI across 1,240+ enterprise documents with 93.6% precision.

Leasel — 8-Agent Enterprise Automation Platform (LangGraph, RAG, HITL)

Agentic AI · Enterprise Automation · Production AI

2026 · Python, LangGraph (8-agent StateGraph), FastAPI, TF-IDF RAG, OCR, SQLite, OpenAI/Anthropic APIs · github.com/venkatalakshmiparimala/Lease_complaince_rag

- Designed and deployed a **production enterprise automation platform**: 8-agent LangGraph StateGraph (Triage → Retrieval → Configuration → Historical Incident → Structured Data → Reasoning → Validation → Human Escalation) performing multi-step reasoning over enterprise documents, settings, and operational data -- with confidence-gated HITL escalation when agent certainty falls below threshold.
- RAG pipeline with TF-IDF retrieval and **100% citation correctness**; ingestion pipeline handling 12+ enterprise document formats (OCR fallback, section-aware chunking); live HUD regulatory API integration; **93.6% precision and 87.9% F1** across 1,240 enterprise documents; 0 safety boundary violations on full adversarial test suite.

Domain-Specific Agentic Automation System (LangGraph, RAG, Compliance Guardrails)

Agentic Automation · Domain AI

2025 · Python, LangGraph, FastAPI, RAG, Pydantic, MAVLink/QGroundControl export

- Built an end-to-end **agentic automation pipeline** that parses natural-language business requests into structured, validated outputs (LangGraph: parse → plan → retrieve → validate → export); RAG compliance layer with cited-source answers over a regulatory corpus; deterministic guardrails blocking non-compliant outputs at every stage before delivery -- demonstrates the full build-validate-deploy cycle for domain-specific agents.

EDUCATION

M.S. Data Analytics Engineering — George Mason University, Virginia — May 2026 · GPA 3.87

B.Tech Computer Science & Engineering — JNTUA, India — 2013–2017

TECHNICAL SKILLS

Agentic AI & LLMs: Agentic AI, Autonomous AI Agents, Multi-Agent Orchestration, LangGraph, LangChain, LlamaIndex, MCP (Model Context Protocol), MCP Servers, LLM Orchestration, Tool Calling, Function Calling, HITL, Structured Outputs, OpenAI API, Anthropic API, Groq, Ollama, Prompt Engineering, Few-Shot, Chain-of-Thought, Guardrails, Hallucination Detection, Confidence Scoring

LLM Evaluation & CI/CD: LLM Evaluation Harness, CI/CD for LLM Systems, LLM-as-Judge, Scenario Simulation, Regression Detection, Golden Datasets, Eval-Driven Development, Agent Observability, Trace-Level Observability, A/B Testing, LLMOps, MLflow, Model Evaluation

RAG & Retrieval: RAG Pipelines, Retrieval-Augmented Generation, Hybrid Search, Semantic Search, Vector Database, ChromaDB, Qdrant, Pinecone, TF-IDF, Embeddings, Document Intelligence, OCR, spaCy, BERT, Hugging Face

Backend & Infrastructure: Python, FastAPI, REST APIs, TypeScript, Node.js, Flask, Java, Spring Boot, Docker, Kubernetes, CI/CD (GitHub Actions), AWS (S3, EC2, Lambda, Glue, Step Functions, Bedrock), Azure (ADF, ADLS, Databricks, AI Foundry), GCP, Terraform, PostgreSQL, SQL, Apache Airflow, Apache Kafka, RabbitMQ, Redis, Microservices, Git, Linux, Agile/Scrum

Enterprise Automation & Integration: Enterprise Workflow Automation, Salesforce, CRM Integration, API Integration, ETL Pipelines, Data Pipelines, Async Processing, Idempotency, Workflow Orchestration, Prefect, Temporal, Celery